Data Architect

Responsible for designing and managing how the workstream's data is structured, stored, integrated, and used to support business operations, reporting, analytics, and decision-making. On this programme, that data includes the clinical Knowledge Graph (KG) and Bayesian Network (BN) artefacts behind the triage solution, so this role also owns the versioned, clinically traceable contracts those artefacts are built from.

Main Responsibilities

1. Design Data Architecture

- Create high-level and detailed data models.
- Define how data is organised across applications and systems.
- Design databases, data warehouses, data lakes, and graph data stores.

2. Data Modelling

- Develop conceptual, logical, and physical data models.
- Define data entities, relationships, and business rules.
- Ensure consistency and standardisation of data structures.
- Model the clinical ontology and Knowledge Graph and align it to SNOMED CT as the primary terminology standard.
- Define and maintain the versioned schema contracts the build pipeline produces and consumes.

3. Data Integration

- Design data flows between systems.
- Define integration methods such as APIs, ETL/ELT, and event-driven architectures.
- Define controlled ingestion paths for clinical source data into AI/ML environments.

4. Data Governance

- Establish data standards and policies.
- Support data quality, metadata, and master data management.
- Ensure data is accurate, consistent, and trusted.
- Own data lineage and provenance as a first-class responsibility, not just a supporting activity.

5. Security and Compliance

- Design data security controls.

- Ensure compliance with regulations such as GDPR, NHS Data Security and Protection Toolkit (DSPT).
- Define data retention, classification, and access controls.
- Support clinical safety governance (DCB0129/DCB0160).
- Confirm data residency and handling compliance for cloud AI/ML workspaces (e.g. SageMaker, Bedrock), working with Security and Information Governance, consistent with existing engineering-workspace decisions.

6. Support Analytics, Reporting, and AI/ML

- Design structures that enable business intelligence and reporting.
- Ensure data is accessible for analytics, AI, and machine learning initiatives.
- Improve data availability and performance.

7. Versioned and Immutable Artefact Governance

- Define and maintain the versioning strategy for immutable, versioned model artefacts.
- Ensure every artefact version can be traced to the source data, ontology version and clinical approval that produced it.
- Define change-control and review process for schema/contract changes, distinct from day-to-day probability recalibration.

8. Stakeholder Collaboration

- Work with client, product teams, engineers, and solution architects.
- Work directly with Clinical Safety, Information Governance, and Data Science/Knowledge Engineering teams.
- Translate business and clinical requirements into data solutions.
- Provide guidance on data-related decisions.

9. Ensure Architectural Compliance

- Review solution designs.
- Ensure projects follow data standards and architecture principles, including DCB0129/DCB0160 clinical safety governance.
- Identify and mitigate data-related risks.